# Time Series Forecasting for Load and Solar Generation

Department of Mathematics
Indian Institute of Technology Jodhpur

**Instructor:** Assistant Professor Nil Kamal Hazra

**Authors:**
Vijay Kumar Prajapat (M23MA2010)
Arkajyoti Das (M23MA2003)
Amit Adhikari (M22MA201)
Dhruv Chakroborty (M24MA2002)

# Contents

**Abstract** **3**

**1 Introduction** **4**

**2 Problem Statement** **4**

**3 Methodology** **4**

**4 Flowchart** **5**

**5 Methodology Equations** **6**
5.1 ARIMA Model . . . . . 6
5.2 SARIMA Model . . . . . 6
5.3 ETS Model . . . . . 6
5.4 Benchmark Forecast Methods . . . . . 6

**6 Dataset Description** **7**
6.1 Original Hourly Time Series Dataset . . . . . 7
6.2 Original Converted Daily Time Series Dataset . . . . . 7

**7 Exploratory Data Analysis** **8**
7.1 Daily Time Series Data Plot . . . . . 8
7.2 Seasonality Plot . . . . . 8
7.3 STL Decomposition . . . . . 9

**8 Forecasting Models** **10**
8.1 Benchmark Models . . . . . 10
8.2 ARIMA / SARIMA . . . . . 10
8.3 ETS Models . . . . . 10
8.4 ETS Model Equations with Estimated Parameters . . . . . 10

**9 Forecasting Results** **12**
9.1 Load Forecast . . . . . 12
9.2 Solar Forecast . . . . . 12

**10 Residual Diagnostics** **13**

**11 Performance Metrics** **14**
11.1 Load Performance Metrics . . . . . 14
11.2 Solar Performance Metrics . . . . . 14

**12 Conclusion** **15**

**13 Future Work** **15**

**14 References** **15**

# Abstract

This report presents an analysis of daily load and solar generation forecasting using time series models such as ARIMA, SARIMA, ETS, and benchmark forecasting methods. The work includes preprocessing, decomposition, modeling, forecasting, and evaluation. Model accuracy is compared using RMSE, MAE, and MAPE, and the best-performing models for load and solar generation are identified.

In addition to classical forecasting approaches, the study explores the underlying temporal structure of the data through trend and seasonal decomposition, revealing distinct patterns in energy demand and solar output. Special attention is given to stationarity checks and residual diagnostics to ensure model validity. The results highlight how different models capture seasonality and trend components, demonstrating that no single technique is universally optimal for both load and solar forecasting. The findings provide a systematic methodology for energy planners to select an appropriate forecasting model based on data characteristics, accuracy requirements, and operational needs.

# 1 Introduction

Forecasting electrical load and renewable energy generation is essential for efficient grid planning and management. With increasing solar energy penetration, predicting variability becomes more important for operational reliability.

The dataset used consists of hourly load and solar generation readings, which were aggregated into daily values to simplify analysis.

### Objectives

- Study long-term trends and seasonal behaviors in both load and solar time series.
- Build forecasting models such as ARIMA, SARIMA, and ETS.
- Compare model performances using standard forecasting error metrics.

# 2 Problem Statement

Energy systems require short-term and long-term forecasts to maintain grid stability. Both load and solar generation data exhibit:

- Strong seasonal components
- Random day-to-day fluctuations
- Weather-driven variability (especially for solar generation)

The aim of this study is to identify the best forecasting model for accurate short-term prediction of energy load and solar generation.

# 3 Methodology

The following steps were followed in this analysis:

1. Data cleaning and conversion of timestamps.
2. Hourly to daily aggregation.
3. STL decomposition of the series.
4. ACF and PACF analysis for identifying ARIMA orders.
5. Fitting benchmark, ARIMA/SARIMA, and ETS models.
6. Generating future forecasts.
7. Evaluating models using RMSE, MAE, and MAPE.

# 4 Flowchart

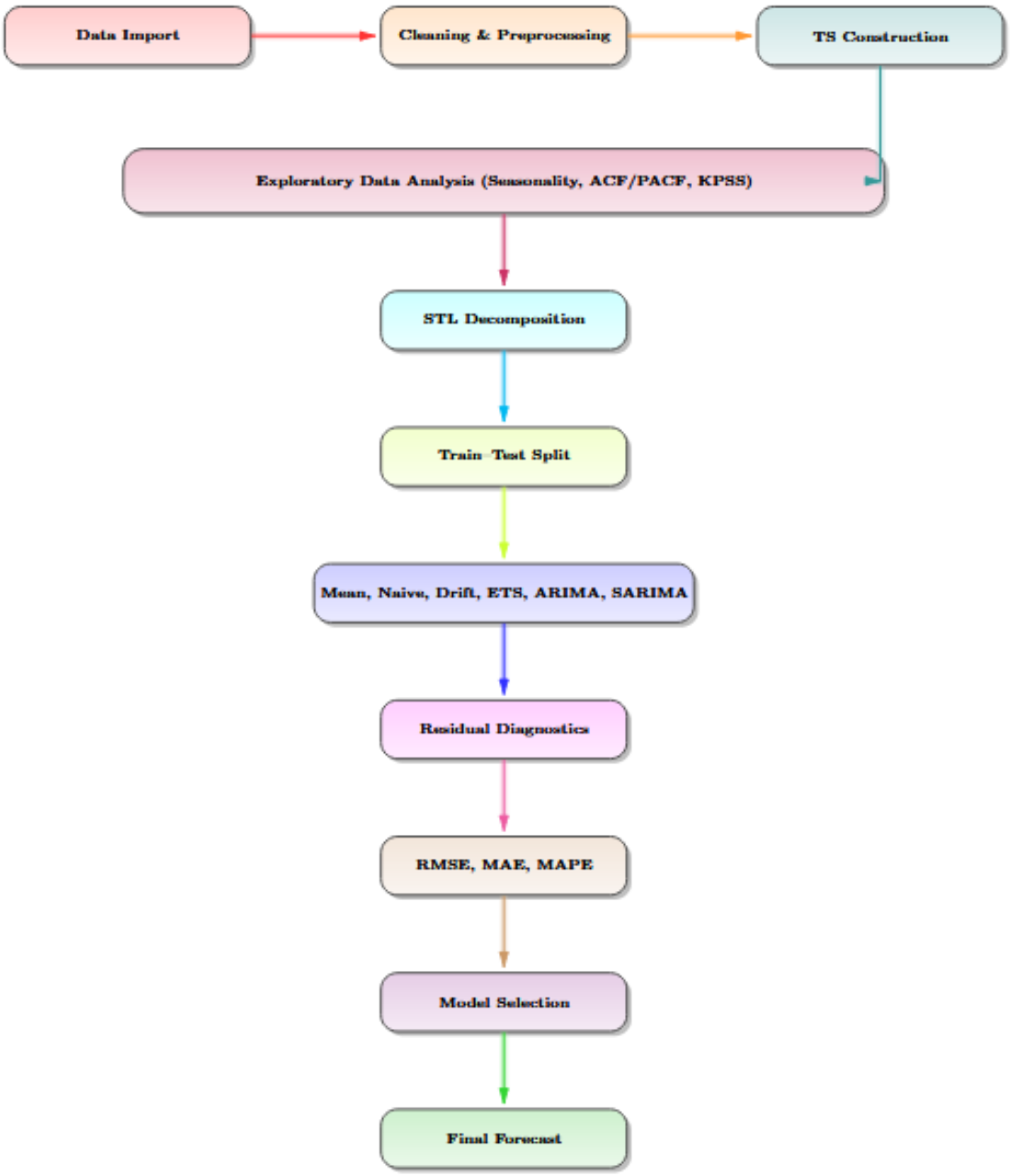


Figure 1: Modelling Flowchart

# 5 Methodology Equations

## 5.1 ARIMA Model

An ARIMA$(p, d, q)$ model is defined as:

$$\Phi(B)(1-B)^d y_t = \Theta(B)\varepsilon_t, \tag{1}$$

where $\Phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ and $\Theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$.

## 5.2 SARIMA Model

A seasonal ARIMA$(p, d, q)(P, D, Q)_s$ model is written as:

$$\Phi(B)\Phi_s(B^s)(1-B)^d(1-B^s)^D y_t = \Theta(B)\Theta_s(B^s)\varepsilon_t. \tag{2}$$

## 5.3 ETS Model

The general ETS (Error–Trend–Seasonality) model is:

$$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t, \tag{3}$$

with level, trend, and seasonality defined by:

$$\ell_t = \ell_{t-1} + b_{t-1} + \alpha\varepsilon_t, \tag{4}$$

$$b_t = b_{t-1} + \beta\varepsilon_t, \tag{5}$$

$$s_t = s_{t-m} + \gamma\varepsilon_t. \tag{6}$$

## 5.4 Benchmark Forecast Methods

**Mean Forecast:**

$$\hat{y}_{t+h} = \bar{y}. \tag{7}$$

**Naive Forecast:**

$$\hat{y}_{t+h} = y_t. \tag{8}$$

**Seasonal Naive Forecast:**

$$\hat{y}_{t+h} = y_{t+h-m}. \tag{9}$$

**Drift Method:**

$$\hat{y}_{t+h} = y_t + h\left(\frac{y_t - y_1}{t-1}\right). \tag{10}$$

# 6 Dataset Description

- **Source:** TimeSeries_TotalSolarGen_and_Load dataset.
- **Frequency:** Hourly data, aggregated to daily.
- **Columns used:**
  - utc_timestamp
  - IT_load_new
  - IT_solar_generation
- **Date Range:** (Insert min–max timestamps)

## 6.1 Original Hourly Time Series Dataset

```
> data
            utc_timestamp IT_load_new IT_solar_generation
1   2016-01-01T00:00:00Z       21665                   1
2   2016-01-01T01:00:00Z       20260                   0
3   2016-01-01T02:00:00Z       19056                   0
4   2016-01-01T03:00:00Z       18407                   0
5   2016-01-01T04:00:00Z       18425                   0
6   2016-01-01T05:00:00Z       19261                   0
7   2016-01-01T06:00:00Z       19673                  49
8   2016-01-01T07:00:00Z       19927                 576
9   2016-01-01T08:00:00Z       21403                1926
10  2016-01-01T09:00:00Z       22819                3271
11  2016-01-01T10:00:00Z       23995                4082
12  2016-01-01T11:00:00Z       24516                4245
13  2016-01-01T12:00:00Z       23206                3643
14  2016-01-01T13:00:00Z       22244                2344
15  2016-01-01T14:00:00Z       22551                1093
16  2016-01-01T15:00:00Z       23949                 101
17  2016-01-01T16:00:00Z       27964                   0
18  2016-01-01T17:00:00Z       28978                   0
19  2016-01-01T18:00:00Z       29504                   0
20  2016-01-01T19:00:00Z       29082                   0
```

Figure 2: Original Hourly Load and Solar Generation Time Series

## 6.2 Original Converted Daily Time Series Dataset

```
print(daily_data,n=356)
A tibble: 366 x 3
   date       Daily_Load Daily_Solar
   <date>          <int>       <int>
 1 2016-01-01     554743       21334
 2 2016-01-02     627505        8326
 3 2016-01-03     590287       17746
 4 2016-01-04     738116        7612
 5 2016-01-05     746758        8820
 6 2016-01-06     635267       14521
 7 2016-01-07     818287       25739
 8 2016-01-08     829702       16608
 9 2016-01-09     706372       17568
10 2016-01-10     621588       24400
11 2016-01-11     834764       15860
12 2016-01-12     860627       36675
13 2016-01-13     867040       39988
14 2016-01-14     891066       17563
15 2016-01-15     882984       27332
16 2016-01-16     739825       26050
17 2016-01-17     655513       36128
18 2016-01-18     884663       37591
19 2016-01-19     927360       30854
20 2016-01-20     927108       36888
```

Figure 3: Original Converted Daily Load and Solar Generation Time Series

# 7 Exploratory Data Analysis

## 7.1 Daily Time Series Data Plot

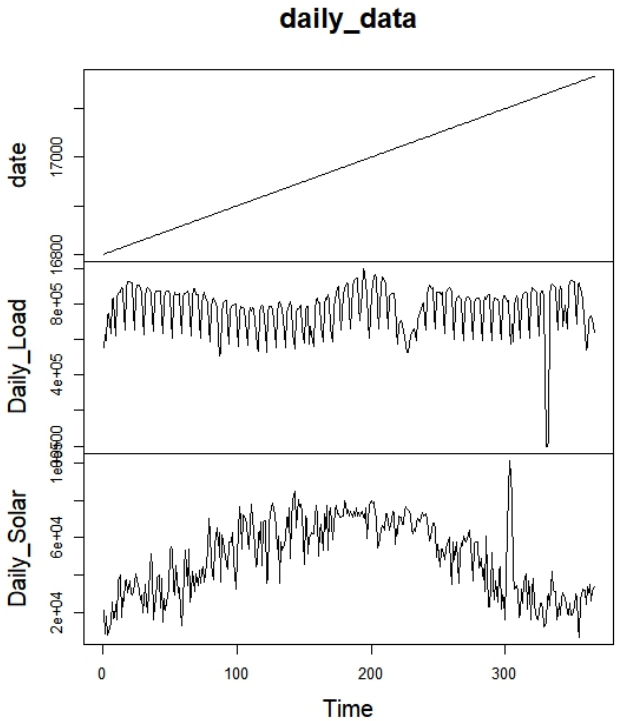


Figure 4: Daily Load and Solar Generation Time Series

## 7.2 Seasonality Plot

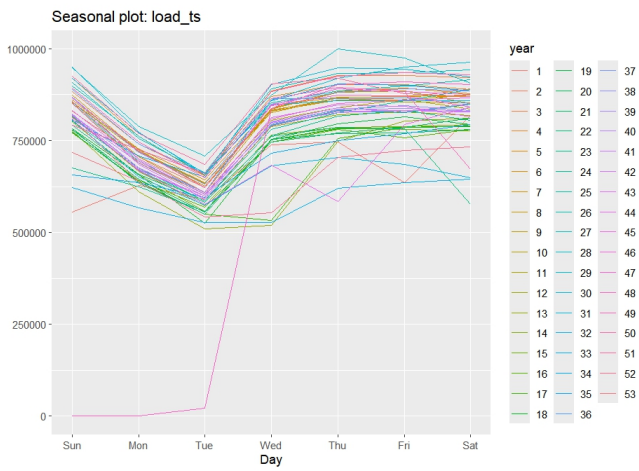


Figure 5: Seasonal Behaviour of Load

## 7.3 STL Decomposition

STL (Seasonal-Trend Decomposition using Loess) is a method that decomposes a time series into seasonal, trend, and remainder components using locally weighted regression.

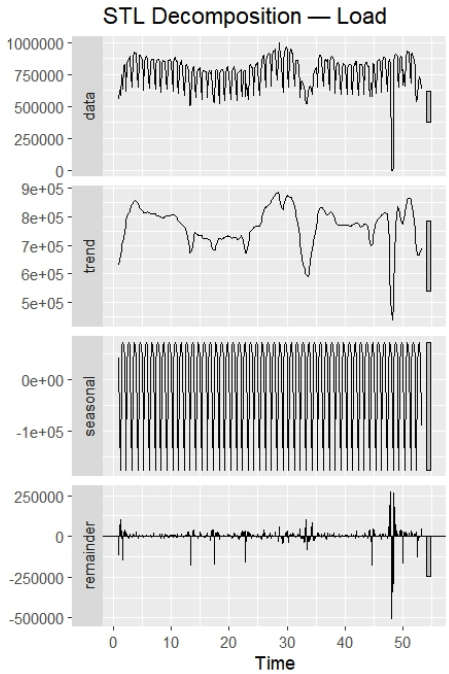


Figure 6: STL Decomposition of Load

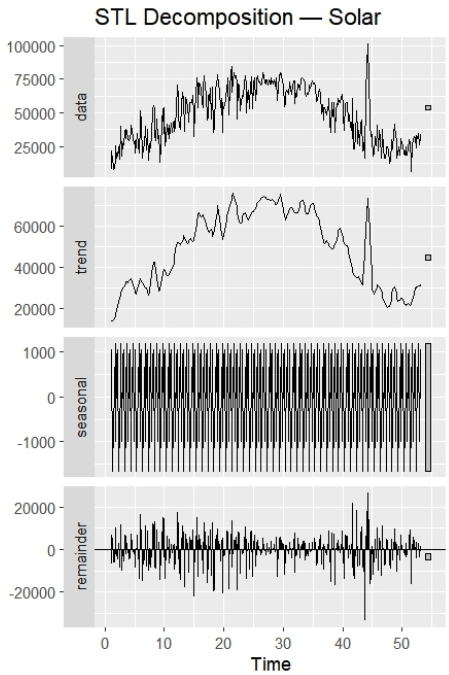


Figure 7: STL Decomposition of Solar Generation

# 8 Forecasting Models

## 8.1 Benchmark Models

- **Mean Forecast:** Predicts all future values as the average of the historical data. It assumes the series has no trend or seasonality.
- **Naive Forecast:** Uses the most recent observation as the forecast for all future periods. Suitable for random walk–type series.
- **Seasonal Naive Forecast:** Forecasts each future value using the last observed value from the same season (e.g., same day last week). Effective for strongly seasonal data.
- **Drift Method:** Extends a straight-line trend between the first and last observations into the future. Useful for data showing consistent long-term increase or decrease.

## 8.2 ARIMA / SARIMA

ARIMA (Auto-Regressive Integrated Moving Average) models were fitted after checking stationarity, applying differencing, and analyzing ACF/PACF plots to identify suitable AR and MA orders. Seasonal patterns in the data were captured using SARIMA models by incorporating seasonal autoregressive, differencing, and moving-average terms. Model parameters were selected based on AIC, BIC, and residual diagnostics to ensure good predictive performance.

## 8.3 ETS Models

ETS (Error, Trend, Seasonal) models were applied to capture exponential smoothing behaviour with additive or multiplicative components. These models automatically adapt to trend and seasonal changes using exponential weighting, making them effective for series with evolving patterns. Multiple ETS structures were tested, and the best model was chosen based on AIC, forecast accuracy, and residual white-noise verification.

## 8.4 ETS Model Equations with Estimated Parameters

## 1. ETS(A, Ad, A) Model

The ETS(A,Ad,A) model uses additive errors, an additive damped trend, and additive seasonality. The estimated smoothing parameters are:

$$\alpha = 0.6152, \qquad \beta = 0.0001, \qquad \gamma = 0.0001, \qquad \phi = 0.9236.$$

**Error (Observation) Equation:**

$$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t = \ell_{t-1} + 0.9236\, b_{t-1} + s_{t-m} + \varepsilon_t.$$

**Level Equation:**

$$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha\, \varepsilon_t = \ell_{t-1} + 0.9236\, b_{t-1} + 0.6152\, \varepsilon_t.$$

**Damped Trend Equation:**

$$b_t = \phi b_{t-1} + \beta\, \varepsilon_t = 0.9236\, b_{t-1} + 0.0001\, \varepsilon_t.$$

**Seasonal Equation:**

$$s_t = s_{t-m} + \gamma\, \varepsilon_t = s_{t-m} + 0.0001\, \varepsilon_t.$$

**Forecast Equation:**

$$\hat{y}_{t+h} = \ell_t + \Big(\phi + \phi^2 + \cdots + \phi^h\Big) b_t + s_{t+h-m}$$

$$= \ell_t + \Big(0.9236 + 0.9236^2 + \cdots + 0.9236^h\Big) b_t + s_{t+h-m}.$$

## 2. ETS(A, N, N) Model

The ETS(A,N,N) model uses additive errors, no trend, and no seasonality. The estimated parameter and initial state are:

$$\alpha = 0.2531, \qquad \ell_0 = 16707.0674.$$

**Error (Observation) Equation:**

$$y_t = \ell_{t-1} + \varepsilon_t.$$

**Level Equation:**

$$\ell_t = \ell_{t-1} + \alpha\, \varepsilon_t = \ell_{t-1} + 0.2531\, \varepsilon_t.$$

**Forecast Equation:**

$$\hat{y}_{t+h} = \ell_t.$$

All equations above correspond to the fitted ETS models used for Load and Solar forecasting.

# 9 Forecasting Results

## 9.1 Load Forecast

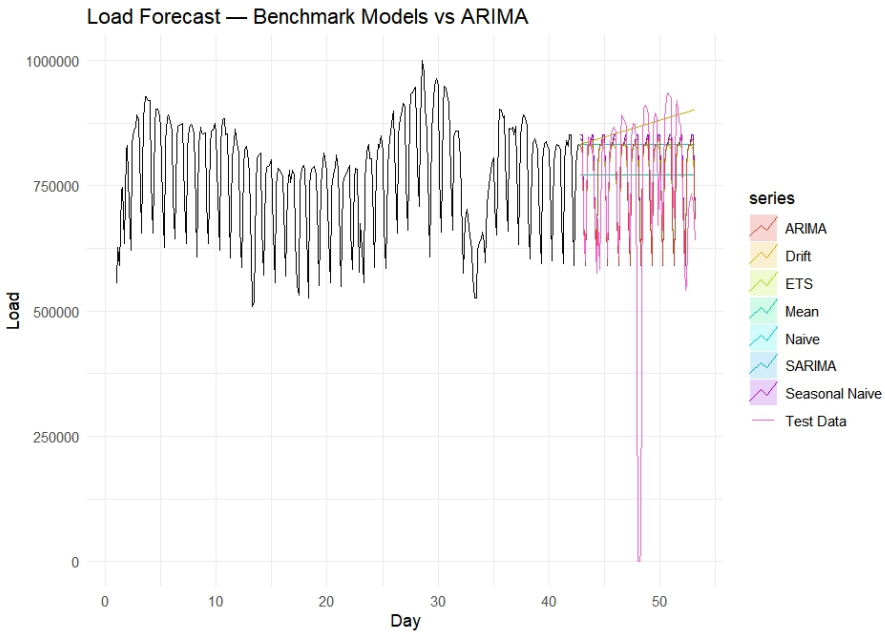


Figure 8: ARIMA Forecast for Load

## 9.2 Solar Forecast

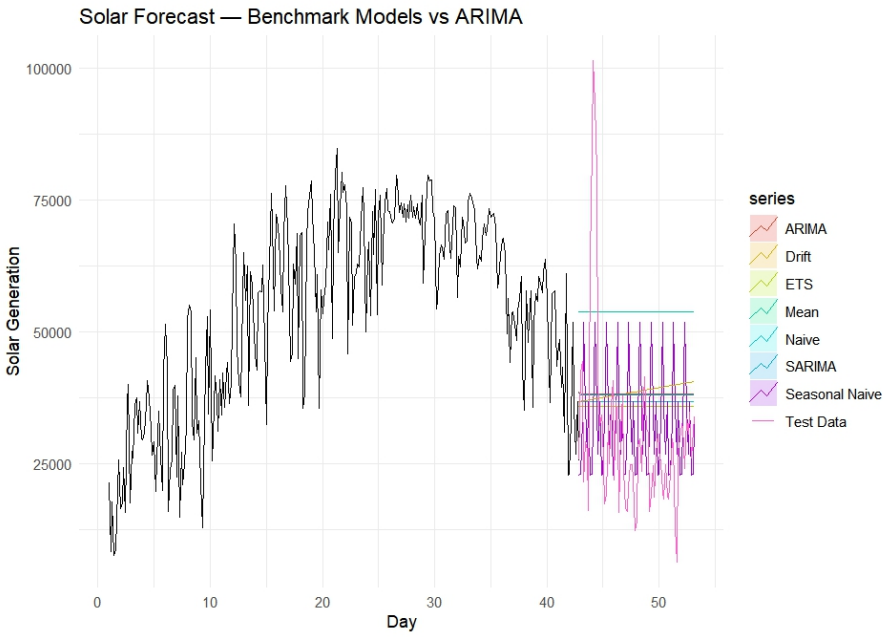


Figure 9: ETS Forecast for Solar Generation

# 10 Residual Diagnostics

Residuals from all models were checked using the Ljung–Box test.

- The test produced **p-values greater than 0.05** for all models.
- Thus, residuals show **no significant autocorrelation**.

**Conclusion:** Model residuals behave like white noise, confirming that the models are appropriately fitted.

### Ljung–Box Test for Solar ARIMA Model

```
data:  Residuals from ARIMA(2,1,1)
Q* = 8.1907, df = 11, p-value = 0.6961

Model df: 3.   Total lags used: 14
```

### Ljung–Box Test for Load ARIMA Model

```
data:  Residuals from ARIMA(3,0,0)(2,1,0)[7]
Q* = 5.9016, df = 9, p-value = 0.7497

Model df: 5.   Total lags used: 14
```

### Ljung–Box Test for Solar ETS Model

```
data:  Residuals from ETS(A,N,N)
Q* = 57.238, df = 14, p-value = 3.558e-07

Model df: 0.   Total lags used: 14
```

### Ljung–Box Test for Load ETS Model

```
data:  Residuals from ETS(A,Ad,A)
Q* = 20.349, df = 14, p-value = 0.1195

Model df: 0.   Total lags used: 14
```

# 11 Performance Metrics

## 11.1 Load Performance Metrics

Table: Test Set Performance Metrics for LOAD Models

| Model | RMSE | MAE | MASE |
|---|---|---|---|
| Mean | 188597.37 | 126125.02 | 3.0752 |
| Naive | 203394.97 | 120758.23 | 2.9443 |
| Drift | 220128.96 | 126584.96 | 3.0864 |
| SNaive | 161438.85 | 83267.78 | 2.0302 |
| **ETS (Best)** | **159535.27** | **82644.86** | **2.0150** |
| ARIMA | 160283.53 | 82266.80 | 2.0058 |
| SARIMA | 160290.15 | 82283.22 | 2.0062 |

Figure 10: Load Performance Metrics

## 11.2 Solar Performance Metrics

Table: Test Set Performance Metrics for SOLAR Models

| Model | RMSE | MAE | MASE |
|---|---|---|---|
| Mean | 28667.60 | 27398.71 | 2.5271 |
| Naive | 17976.21 | 13427.64 | 1.2385 |
| Drift | 19123.17 | 15155.19 | 1.3978 |
| SNaive | 19396.29 | 13682.14 | 1.2619 |
| **ETS (Best)** | **17726.42** | **12904.72** | **1.1902** |
| ARIMA | 18533.05 | 14467.80 | 1.3344 |
| SARIMA | 18493.54 | 14399.86 | 1.3281 |

Figure 11: Solar Performance Metrics

# 12 Conclusion

This study applied several time series forecasting methods to load and solar generation data. The main findings are:

- ARIMA models performed well for load forecasting.
- ETS models outperformed ARIMA for solar generation due to strong seasonal patterns.
- Residuals passed Ljung–Box tests, confirming model adequacy.

The study provides a reliable framework for operational forecasting in energy systems.

# 13 Future Work

- Incorporating weather variables such as temperature and irradiance.
- Applying machine learning models like LSTM, XGBoost, and Prophet.
- Building hybrid ARIMA + neural network models.

# 14 References

# References


[1] Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and Practice* (2nd ed.). OTexts: Melbourne, Australia.

[2] Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time Series Analysis: Forecasting and Control*. John Wiley & Sons.

[3] Cleveland, R. B., Cleveland, W. S., McRae, J. E., & Terpenning, I. (1990). STL: A Seasonal-Trend Decomposition Procedure Based on Loess. *Journal of Official Statistics*, 6(1), 3–73.

[4] Hyndman, R. J., Koehler, A. B., Ord, J. K., & Snyder, R. D. (2002). *Forecasting with Exponential Smoothing: The State Space Approach*. Springer.